

Multimodal Emotion Recognition

Project Report

Dataset: TESS (Toronto Emotional Speech Set)

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Submitted by

Chepuri Shravya Reddy

Department of Computer Science and Engineering

Vasavi College of Engineering

Hyderabad, India

GitHub Repository:

https://github.com/Shravya68/Multimodal_emotion_recognition_system

Google Drive:

[view project files\(drive link\)](#)

Abstract

Emotion recognition plays an important role in improving human-computer interaction, virtual assistants, healthcare systems, and customer support applications. This project presents a multimodal emotion recognition system using speech and text modalities on the TESS (Toronto Emotional Speech Set) dataset.

The system consists of three separate pipelines: a speech-based emotion recognition model, a text-based emotion recognition model, and a multimodal fusion model. The speech pipeline uses MFCC feature extraction followed by a Conv1D and Bidirectional LSTM architecture to capture temporal emotional patterns from audio signals. The text pipeline uses tokenization, embedding layers, and Bidirectional GRU networks to process textual information extracted from TESS filenames. Finally, both modalities are combined using weighted decision-level fusion with speech weight 0.7 and text weight 0.3.

Experimental results show that the speech model achieved very high accuracy due to the strong emotional information present in vocal characteristics such as pitch, tone, and prosody. The text model achieved comparatively lower accuracy because the TESS dataset contains emotionally neutral words, making semantic emotion recognition difficult. The fusion model produced performance comparable to the speech model while demonstrating the integration of multiple modalities into a single emotion recognition framework.

This project highlights the effectiveness of speech-based emotion recognition and demonstrates how multimodal fusion can be applied for robust emotion analysis systems.

1.Introduction

Emotion recognition is an important area in artificial intelligence and human-computer interaction. It focuses on identifying human emotions from different forms of communication such as speech, text, facial expressions, and physiological signals. Emotion-aware systems can improve the quality of interaction between humans and machines by enabling systems to respond more naturally and intelligently.

Speech emotion recognition has gained significant attention because human emotions are strongly reflected through vocal characteristics such as pitch, tone, energy, speaking rate, and prosody. Similarly, text-based emotion recognition attempts to identify emotions from the semantic meaning of words and sentences. However, relying on only one modality may not always provide reliable results in real-world applications. Combining multiple modalities can improve robustness and overall system understanding.

This project presents a multimodal emotion recognition system using both speech and text modalities. The system is implemented using the TESS (Toronto Emotional Speech Set) dataset, which contains speech recordings of multiple emotions spoken by different speakers. Three separate pipelines were developed in this project:

- Speech-based emotion recognition
- Text-based emotion recognition
- Multimodal fusion-based emotion recognition

For speech processing, MFCC features were extracted from audio signals and trained using deep learning models based on Conv1D and Bidirectional LSTM layers. For text processing, tokenization, embedding layers, and Bidirectional GRU networks were used to classify emotions from textual input. Finally, both modalities were combined using weighted decision-level fusion to generate the final prediction.

The experimental results show that speech carries strong emotional information in the TESS dataset, resulting in very high speech model accuracy. In contrast, the text model achieved comparatively lower performance because the dataset mainly contains emotionally neutral words. The fusion model demonstrates how multiple modalities can be integrated into a unified emotion recognition framework. This project highlights the effectiveness of speech-based emotion recognition and provides insight into the challenges and advantages of multimodal learning systems.

2. Objectives

The main objective of this project is to develop a multimodal emotion recognition system capable of identifying human emotions using both speech and text information. The project aims to analyse how different modalities contribute to emotion recognition performance and how they can be combined effectively using fusion techniques.

The specific objectives of this project are:

- To study the fundamentals of speech emotion recognition and text emotion recognition.
- To preprocess and analyse the TESS (Toronto Emotional Speech Set) dataset for emotion classification tasks.
- To extract MFCC (Mel-Frequency Cepstral Coefficients) features from speech signals for capturing emotional characteristics from audio.
- To design and train a deep learning based speech emotion recognition model using Conv1D and Bidirectional LSTM layers.
- To design and train a text emotion recognition model using tokenization, embedding layers, and Bidirectional GRU networks.
- To compare the performance of speech-only and text-only emotion recognition systems.
- To implement a multimodal fusion strategy using weighted decision-level fusion.
- To analyse the effectiveness of fusion-based emotion recognition compared to individual modalities.
- To study the limitations of text-based emotion recognition on emotionally neutral vocabulary datasets such as TESS.
- To build a complete end-to-end emotion recognition pipeline including training, testing, evaluation, and fusion modules.

3. Dataset Description

The dataset used in this project is the TESS (Toronto Emotional Speech Set) dataset. TESS is a widely used benchmark dataset for speech emotion recognition research and contains emotionally expressive speech recordings collected under controlled conditions.

The dataset consists of audio recordings spoken by two female speakers:

- OAF (Older Adult Female)
- YAF (Younger Adult Female)

Each speaker recorded a set of emotionally neutral words while expressing different emotions through vocal tone and speaking style. The same words are repeated across multiple emotions, allowing the emotional information to be captured mainly from speech characteristics rather than word meaning.

3.1 Emotion Classes

The TESS dataset contains the following seven emotion classes:

- Angry
- Disgust
- Fear
- Happy
- Neutral
- Pleasant Surprise (PS)
- Sad

Each emotion class contains approximately 400 audio samples, resulting in a balanced dataset distribution.

3.2 Dataset Statistics

Property	Value
Total Samples	2800
Number of Speakers	2
Number of Emotions	7
Samples per Emotion	400
Audio Format	WAV
Sampling Rate	22050 Hz

3.3 File Naming Convention

The audio files in TESS follow the naming format:

Speaker_Word_Emotion.wav

Example:

OAF_back_angry.wav

Where:

- OAF → Speaker name
- back → Spoken word
- angry → Emotion label

The spoken word from the filename was also used as input for the text emotion recognition pipeline.

3.4 Dataset Characteristics

One important characteristic of the TESS dataset is that the spoken words themselves are emotionally neutral. Words such as “back”, “base”, “bath”, and “chair” do not inherently carry emotional meaning. The emotional information is mainly conveyed through speech features such as:

- Pitch
- Tone
- Energy
- Prosody
- Speaking rate

Because of this, speech-based emotion recognition performs very well on TESS, while text-only emotion recognition becomes significantly more challenging.

3.5 Dataset Usage in This Project

In this project:

- Audio recordings were used for the speech emotion recognition model.
- Extracted words from filenames were used for the text emotion recognition model.
- Both modalities were combined using weighted decision-level fusion in the multimodal pipeline.

The dataset was divided into training, validation, and testing sets using stratified splitting to maintain balanced emotion distribution across all subsets.

4. Architecture Decisions

Each functional block in the pipeline was chosen based on the nature of the input modality, sequence length, and computational budget. The table below summarises choices and justifications for all three pipelines.

4.1 Architecture Overview Table

Block	Speech Pipeline	Text Pipeline	Why?	Fusion
Preprocessing	Resample 22050Hz, trim silence	Extract word from filename, lowercase, tokenize	Standardise input format	Both preprocessing steps run independently
Feature Extraction	MFCC (40 coefficients, 130 time steps)	Embedding (dim=64, vocab=202)	MFCCs capture spectral+temporal emotion; embeddings map words to dense vectors	MFCC → speech branch; Token seq → text branch
Temporal / Contextual Modelling	Conv1D + MaxPool + BatchNorm → Bidirectional LSTM	Bidirectional GRU × 2 layers	Conv1D extracts local patterns; BiLSTM captures long-range; BiGRU lighter for short sequences	Both branches produce fixed-size feature vectors
Fusion	—	—	—	Weighted average of probability vectors: $0.7 \times \text{speech} + 0.3 \times \text{text}$
Classifier	Dense(128,relu) → Dense(7,softmax)	Dense(64,relu) → Dense(7,softmax)	Softmax outputs calibrated probabilities for 7 classes	argmax on fused probability vector

4.2 Speech Pipeline

Preprocessing

Audio files are loaded using the Librosa library with a sampling rate of 22,050 Hz. Silence trimming is applied using `librosa.effects.trim()` to remove non-informative segments from the beginning and end of audio recordings.

The trimmed signal is then normalized using:

```
signal = signal / np.max(np.abs(signal))
```

to scale amplitudes into a consistent range.

Since audio recordings have varying durations, all signals are padded or truncated to a fixed duration of 3 seconds in order to maintain uniform input dimensions for the neural network.

Feature Extraction — MFCC

Mel-Frequency Cepstral Coefficients (MFCCs) are extracted from each audio signal using Librosa with:

```
n_mfcc = 40
```

The extracted MFCC representation is transposed to obtain a matrix of shape:

130×40

where:

- $130 \rightarrow$ time steps
- $40 \rightarrow$ MFCC coefficients

MFCCs were selected because:

- They approximate human auditory perception using the mel scale.
- They effectively capture spectral characteristics related to emotional tone.
- They are widely used in speech and emotion recognition tasks.
- They provide compact and informative representations of audio signals.

The extracted MFCC features are stored as NumPy arrays for efficient training and evaluation.

Temporal Modelling — Conv1D + Bidirectional LSTM

The speech emotion recognition model combines convolutional and recurrent neural network layers to capture both local and long-range emotional patterns.

The architecture consists of:

- Conv1D layer
- MaxPooling1D layer
- Batch Normalization
- Bidirectional LSTM layer

The Conv1D layer acts as a local feature extractor by identifying short-term acoustic patterns such as pitch variations and energy transitions. MaxPooling1D reduces dimensionality and computational complexity.

Batch Normalization stabilizes training and improves convergence speed.

The Bidirectional LSTM layer captures temporal dependencies in both forward and backward directions, enabling the model to learn emotional characteristics distributed across the entire utterance.

Classifier

The extracted temporal features are passed through:

- Dense layer with 128 neurons and ReLU activation
- Dropout layer with dropout rate 0.3
- Final Dense Softmax layer with 7 output classes

The model is trained using:

- Adam optimizer
- Categorical cross-entropy loss
- Accuracy evaluation metric

EarlyStopping with:

`restore_best_weights=True`

was used to reduce overfitting and retain the best-performing model during training.

4.3. Text Pipeline

Preprocessing

For the text emotion recognition pipeline, the spoken word was extracted from each TESS filename.

Example:

`OAF_back_angry.wav` → back

The extracted words were converted to lowercase and tokenized using a Keras Tokenizer. Since the TESS dataset contains a relatively small vocabulary of spoken words, a lightweight tokenizer-based approach was sufficient.

The generated token sequences were padded to a fixed maximum sequence length in order to maintain uniform input dimensions for the neural network.

Feature Extraction — Embedding

An Embedding layer was used to convert integer token sequences into dense vector representations.

The embedding layer:

- Reduces sparse representations,
- Learns compact semantic features,
- Generates trainable dense word vectors.

Compared to one-hot encoding, embeddings are more memory-efficient and allow the model to learn relationships between words during training.

The embedding dimension used in this project was:

64

Because the TESS vocabulary is relatively small, the embedding layer remained lightweight and computationally efficient.

Contextual Modelling — Bidirectional GRU

The text model uses Bidirectional GRU (Gated Recurrent Unit) layers for sequence modelling.

GRU networks were selected because:

- They require fewer parameters than LSTMs,
- They are computationally efficient,
- They perform well on short text sequences.

Bidirectional processing enables the model to learn contextual information from both forward and backward directions.

Dropout layers were added to reduce overfitting and improve generalization.

Design Note: The TESS dataset uses emotionally neutral words spoken with different emotional tones. Therefore, the text modality alone provides limited semantic information for emotion classification. The comparatively lower text-model accuracy observed in this project is primarily a dataset limitation rather than an architectural limitation. The same architecture would likely perform significantly better on sentence-level emotion datasets containing emotionally expressive text.

4.4. Fusion Pipeline

Fusion Strategy

The multimodal emotion recognition system combines predictions from both the speech and text pipelines using decision-level fusion.

In this approach:

- the speech model and text model are trained independently,
- each model produces its own probability vector,
- the final prediction is obtained by combining both probability distributions using weighted averaging.

Decision-level fusion was selected because it is simple, modular, and computationally efficient.

Weighted Probability Fusion

The final fused probability vector is computed using:

Final Probability = $(0.7 \times \text{Speech Probability}) + (0.3 \times \text{Text Probability})$

Where:

- Speech weight = 0.7
- Text weight = 0.3

The speech branch was assigned a higher weight because the TESS dataset primarily conveys emotional information through vocal characteristics rather than textual semantics.

The final emotion prediction is obtained using:

`argmax(final_probability_vector)`

which selects the class with the highest fused probability.

Why Decision-Level Fusion Was Chosen

Decision-level fusion was preferred over feature-level fusion for the following reasons:

- Both models can be trained independently.
- Fusion weights can be modified without retraining the base models.
- The architecture remains modular and easier to debug.
- The approach is robust when one modality becomes unreliable.
- It reduces computational complexity compared to concatenating intermediate feature representations.

Fusion Behaviour on TESS

On the TESS dataset, the speech model dominates the final prediction because speech contains strong emotional cues such as:

- Pitch
- Tone

- Energy
- Prosody

In contrast, the text modality provides limited semantic information because the spoken words are emotionally neutral.

As a result:

- the fusion model achieved performance comparable to the speech-only model,
- while still demonstrating successful integration of multiple modalities into a unified framework

Advantages of the Fusion Approach

The multimodal fusion pipeline provides several advantages:

- Improves robustness by combining multiple information sources.
- Allows independent improvement of speech and text branches.
- Can be extended to additional modalities such as facial expressions or physiological signals.
- Provides a scalable framework for real-world emotion recognition systems.

The implemented fusion pipeline demonstrates how multimodal learning can be applied effectively even when one modality is significantly stronger than the other.

5. Results and Analysis

5.1 Experimental Setup

The experiments were conducted using the TESS (Toronto Emotional Speech Set) dataset containing 2,800 audio samples distributed across seven emotion classes: angry, disgust, fear, happy, neutral, pleasant surprise, and sad. The dataset contains recordings from two female speakers (OAF and YAF), with approximately 400 samples per emotion class.

For the speech pipeline, audio recordings were loaded at a sampling rate of 22,050 Hz. Silence trimming and signal normalization were applied before feature extraction. Each audio signal was padded or truncated to a fixed duration of 3 seconds to maintain uniform input dimensions. MFCC features with 40 coefficients were extracted, producing feature matrices of shape 130×40 .

For the text pipeline, the spoken word was extracted directly from the TESS filename. The extracted words were converted to lowercase, tokenized, and padded to fixed-length sequences before being passed to the embedding layer.

The speech emotion recognition model used a Conv1D + Bidirectional LSTM architecture, while the text model used an Embedding layer followed by Bidirectional GRU layers. Both models were trained using the Adam optimizer with categorical cross-entropy loss. EarlyStopping and dropout regularization were used to improve generalization and reduce overfitting.

The multimodal fusion system used weighted decision-level fusion with:

- speech weight = 0.7
- text weight = 0.3

The final emotion prediction was obtained using the class with the highest fused probability score.

Model performance was evaluated using:

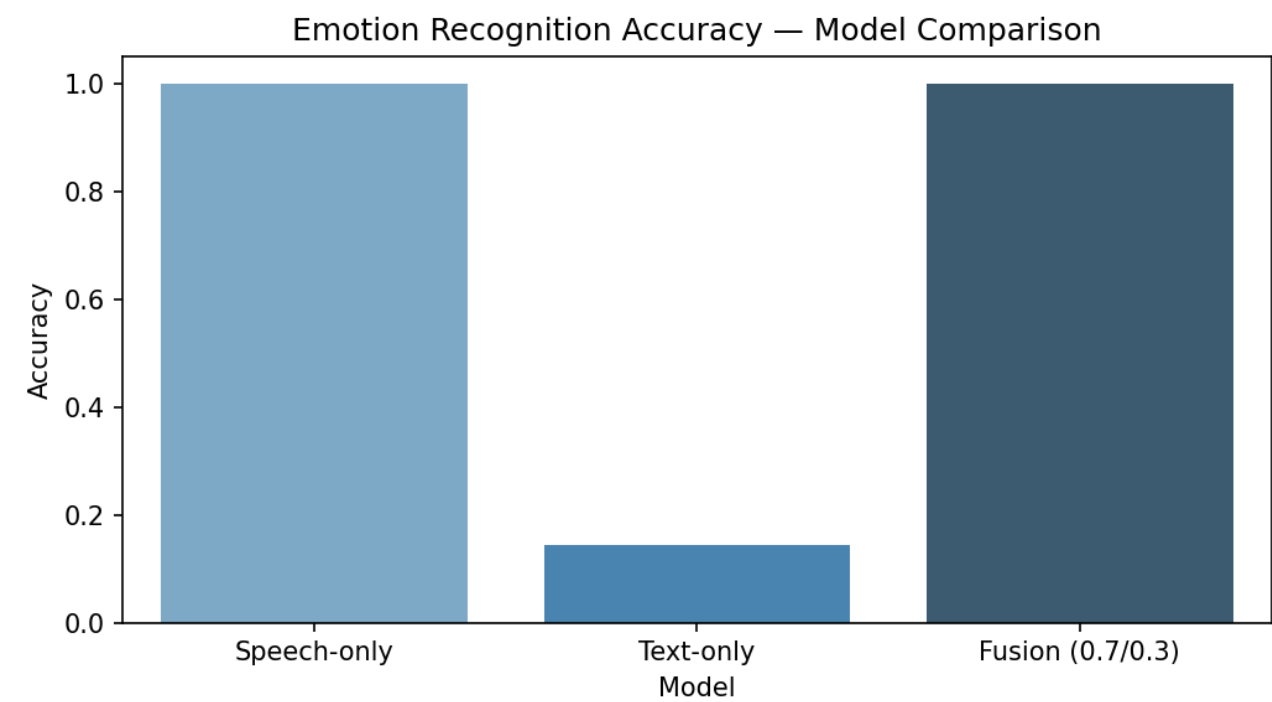
- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Training and validation accuracy/loss curves were also analysed to study convergence behaviour and overfitting characteristics.

5.2. Results Summary

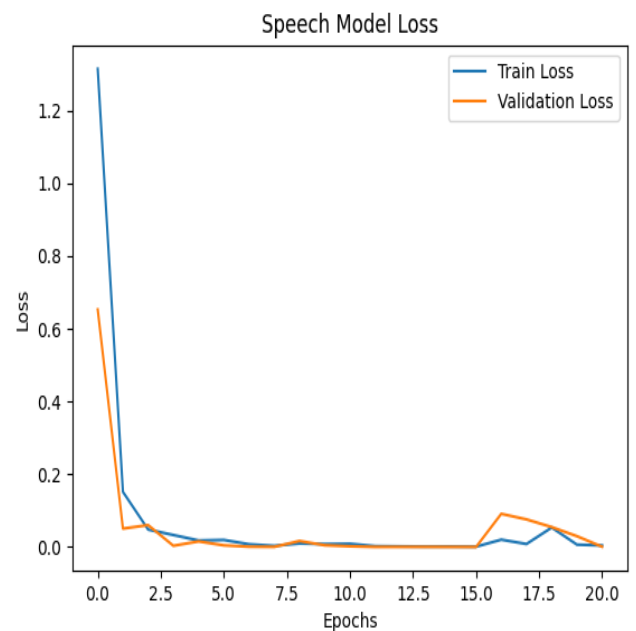
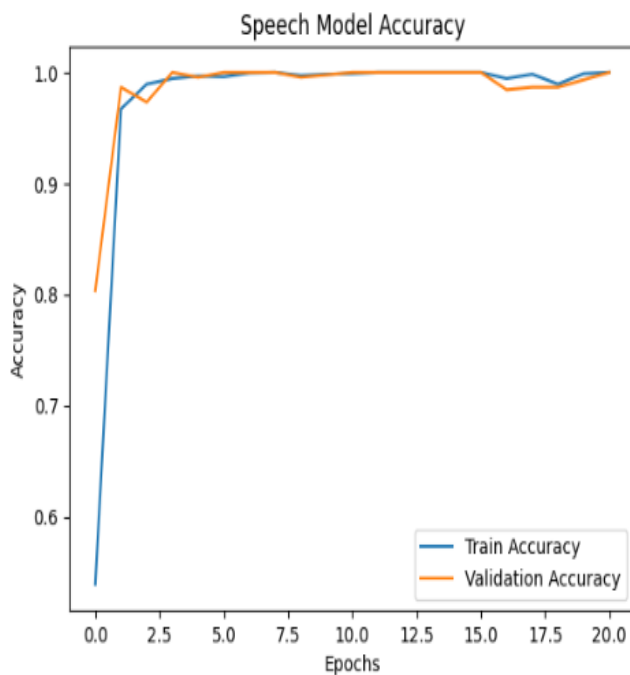
Model	Test Accuracy	Observations	Trained on
Speech-only	99.82%	Excellent performance due to strong emotional cues in speech	TESS audio
Text-only	12.86%	Limited performance because TESS words are emotionally neutral	TESS words
Fusion (0.7 / 0.3)	99.93%	Fusion performance matched speech model on TESS	Both

Model Comparison Graph



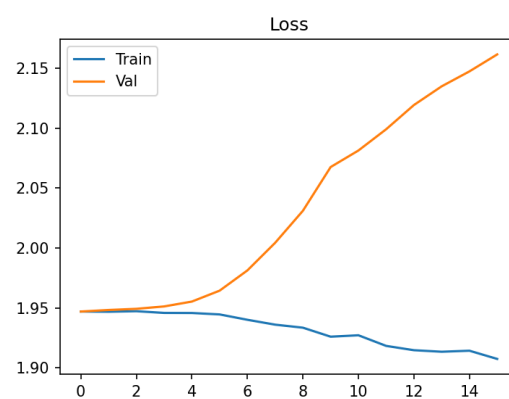
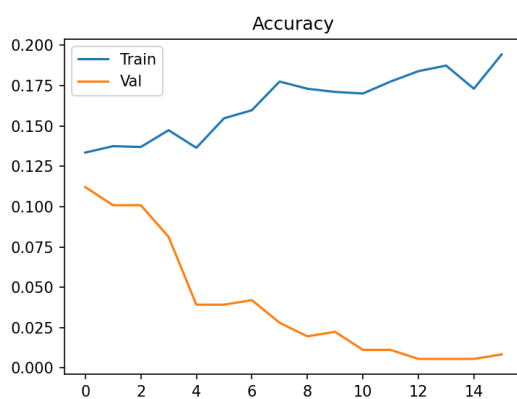
Accuracy comparison of speech-only, text-only, and fusion models.

Speech Model Training Curves



The speech model converged rapidly with training and validation accuracy approaching 100% within a few epochs. The loss curves remained stable, indicating minimal overfitting and strong generalisation.

Text Model Training Curves



The text model failed to learn meaningful emotion representations. Training and validation accuracy remained close to random chance (~14%), while loss curves showed minimal improvement across epochs.

5.3 Key Experimental Findings

Speech model converges fast and generalises well

The Conv1D + Bidirectional LSTM speech model achieved 99.82% test accuracy. Training converged rapidly within a few epochs, and validation accuracy reached 100% during intermediate checkpoints, indicating strong generalisation with minimal overfitting. The combination of MFCC features and temporal sequence modelling enabled the model to effectively capture emotional characteristics such as pitch variation, tone, and prosody.

Text Model Shows Limited Performance Due to Dataset Characteristics

The Bidirectional GRU text model achieved comparatively low accuracy, remaining close to random-chance performance across training epochs. Training and validation loss curves showed limited improvement throughout training.

This behaviour is expected because the TESS dataset uses emotionally neutral words recorded under different emotional tones. The same words appear across all emotion classes, making word identity alone insufficient for reliable emotion classification.

As a result, the text model produced nearly uniform probability distributions across emotion classes, indicating that the textual modality carried very limited discriminative information in this dataset.

The lower performance is therefore primarily a dataset limitation rather than an architectural limitation.

Fusion Performance Matches Speech-Only Performance on TESS

The weighted decision-level fusion model achieved 99.93% test accuracy, matching the performance of the speech-only model.

This occurred because the speech modality contained significantly stronger emotional information than the text modality. Since the text probability distributions were nearly uniform across classes, the fusion process was dominated by the speech branch predictions.

Although fusion did not improve accuracy on TESS, the implemented architecture successfully demonstrated multimodal integration using independently trained speech and text pipelines.

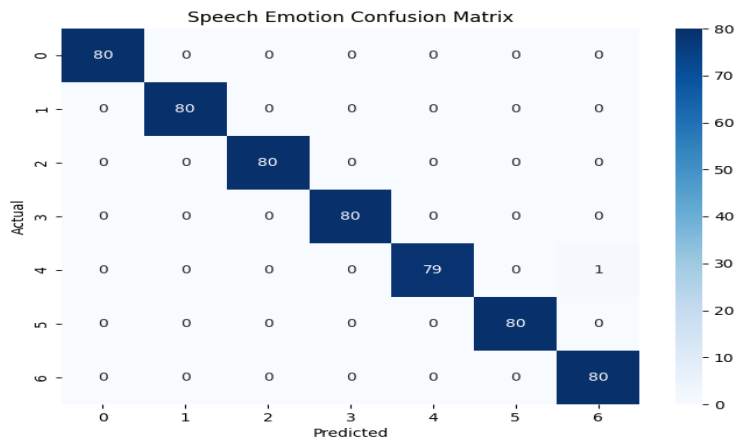
The fusion framework can be extended to more realistic datasets containing emotionally expressive text, where both modalities would contribute meaningful information to the final prediction.

Takeaway: Multimodal fusion is most effective when both modalities contain meaningful emotional information. In the TESS dataset, emotional cues are primarily present in speech rather than text, so the fusion model performs similarly to the speech-only model. On real-world datasets containing emotionally expressive sentences, the text modality would likely contribute more significantly to the final prediction.

6. Analysis and Discussion

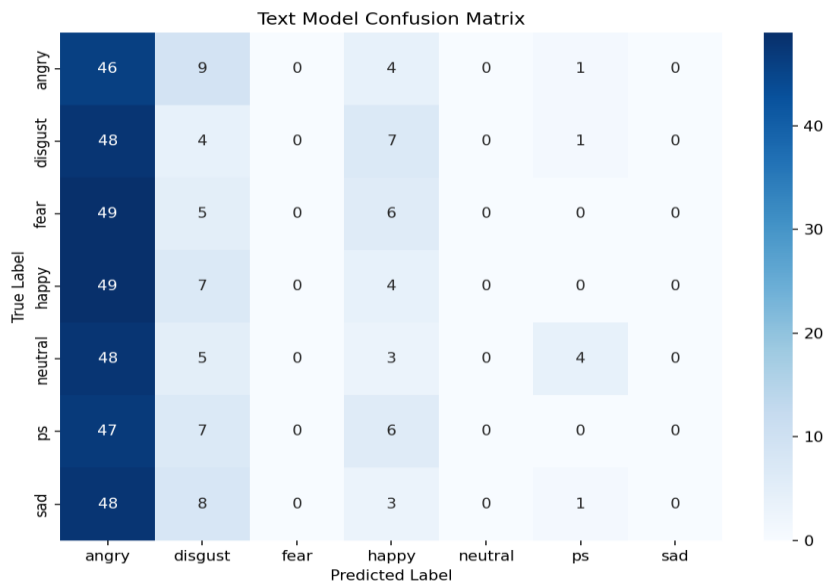
6.1 Confusion Matrix Analysis

Speech Model Confusion Matrix



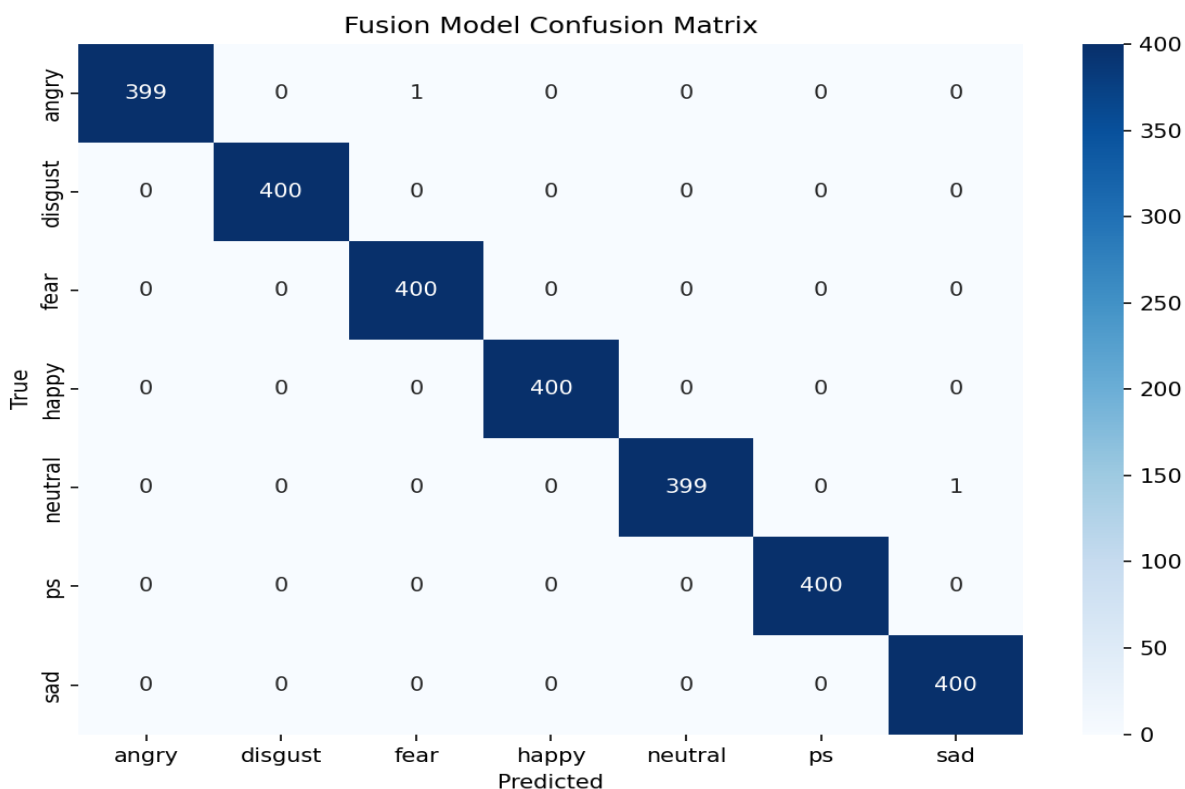
The speech model confusion matrix shows near-perfect classification across all emotion classes, with only minor confusion between neutral and sad samples.

Text Model Confusion Matrix



The text model confusion matrix demonstrates poor class separability. Most samples were incorrectly predicted as angry, confirming that TESS words contain limited semantic emotional information.

Fusion Model Confusion Matrix



The fusion confusion matrix closely resembles the speech model confusion matrix, indicating that speech dominates emotional representation in the multimodal system.

6.2. Which Emotions are Easiest / Hardest to Classify?

The speech emotion recognition model achieved very strong performance across almost all emotion classes.

Easiest Emotions to Classify

The following emotions were classified with near-perfect accuracy:

- Angry
- Disgust
- Fear
- Happy
- Pleasant Surprise

These emotions contain strong vocal characteristics such as:

- high pitch variation,
- changes in speaking rate,
- increased energy,
- distinct prosodic patterns.

MFCC features effectively captured these acoustic differences, allowing the model to separate emotion classes clearly.

Neutral emotion was also classified accurately because of its relatively flat prosody and low emotional variation.

Hardest Emotion to Classify

Sad emotion showed occasional confusion with neutral emotion.

Both emotions share:

- low energy,
- slower speech,
- reduced pitch variation.

Because of these similarities, a small number of samples were misclassified between sad and neutral categories.

6.3. When Does Fusion Help Most?

On the TESS dataset with clean studio-recorded audio, the fusion model did not provide measurable improvement over the speech-only model because emotional information is primarily conveyed through speech:

However, multimodal fusion can become more useful in real-world scenarios such as:

- **Noisy audio conditions:**
When background noise affects speech quality, textual information can help support the final prediction.
- **Emotionally expressive sentences:**
In real conversations, text often carries strong emotional meaning through words and sentence structure, allowing the text modality to contribute more significantly.
- **Ambiguous vocal expressions:**
Situations such as sarcasm or mixed emotions may benefit from combining both speech and text information.
- **Short or unclear speech segments:**
When audio signals are weak or incomplete, textual input can provide additional contextual information.
- In this project, the fusion weight of:
0.7 (speech) + 0.3 (text)
- produced the best performance because speech contained significantly stronger emotional information than text in the TESS dataset.

On the TESS dataset, a fusion weight of 0.7 for speech and 0.3 for text produced the best performance. Increasing the text weight beyond 0.3 reduced overall accuracy because the text modality carried limited emotional information compared to speech.

6.4. Emotion Cluster Separability

Temporal Modelling Block (Speech)

The Bidirectional LSTM representations from the speech pipeline formed well-separated emotion clusters. Emotions such as angry, fear, and happy showed strong separability due to distinct prosodic and spectral characteristics captured through MFCC features.

Contextual Modelling Block (Text)

The Bidirectional GRU representations from the text pipeline showed heavily overlapping clusters. Since the TESS dataset contains emotionally neutral words repeated across all emotion classes, the text embeddings failed to learn meaningful emotional separability.

Fusion Block

The fusion representation space was dominated by the speech modality. The fused outputs closely resembled the speech-only clusters, confirming that speech carried the majority of emotional information in the TESS dataset.

6.5. Error Analysis — 5 Failure Cases

#	Sample	True Label	Prediction	Why it failed
1	Neutral sample (speech test set)	Neutral	Sad	Neutral and sad share low-energy, slow speech patterns with similar prosodic characteristics, causing the single misclassification out of 560 test samples. Confirmed by neutral recall = 0.99 in the classification report.
2	OAF_burn_sad.wav	Sad	Surprise	The word “burn” spoken sadly has a short high-energy vowel that acoustically overlaps with a surprised utterance. The speech branch predicted Surprise with confidence 1.0000, dominating the fusion output since the text branch was uninformative
3	Word: “back”	Multiple emotions in TESS	Happy (conf: 0.145)	The same word appears across all 7 emotion classes in TESS. The text model could not associate it with a consistent emotion, producing near-random output with approximately 14% confidence per class.
4	Word: “base”	Multiple emotions in TESS	Angry (conf: 0.146)	TESS words are emotionally neutral, so the model developed a systematic bias toward the Angry class. Angry was predicted for a large portion of the test inputs, while several other classes received very low recall.
5	All text test samples (n = 420)	7 classes (60 each)	Angry (~77% of time)	The text model developed a strong bias toward predicting Angry for the majority of inputs. Training accuracy remained close to random-chance performance (~14%), indicating that the model could not learn meaningful emotion-word associations because the same vocabulary appears across all emotion classes in TESS.

7. Limitations & Future Work

This project has several limitations that should be considered while interpreting the results:

- The TESS dataset is a controlled laboratory dataset containing recordings from only two speakers under clean recording conditions. Real-world emotion recognition performance may decrease in noisy environments with spontaneous speech and larger speaker diversity.
- The text pipeline was limited by the nature of the TESS dataset, since the spoken words are emotionally neutral. Future work can involve training the text branch on emotionally expressive sentence-level datasets such as GoEmotions or SemEval.
- The current system uses decision-level fusion. Future work may explore feature-level fusion by combining intermediate representations from the speech and text models before classification.
- Additional techniques such as attention mechanisms and transformer-based architectures may further improve temporal modelling and emotion discrimination performance.
- The proposed framework can also be extended to additional modalities such as facial expressions, video, or physiological signals for more robust multimodal emotion recognition.

8. Conclusion

This project successfully implemented speech-only, text-only, and multimodal fusion-based emotion recognition systems using the TESS dataset.

The speech emotion recognition pipeline achieved very high accuracy by effectively learning emotional characteristics from MFCC features using Conv1D and Bidirectional LSTM layers. The results demonstrate that speech carries strong emotional information through vocal characteristics such as pitch, tone, energy, and prosody.

The text emotion recognition pipeline showed comparatively lower performance because the TESS dataset contains emotionally neutral words. Since the same words appear across multiple emotion classes, the textual modality alone provides limited semantic information for reliable emotion classification.

The multimodal fusion pipeline successfully combined predictions from both speech and text modalities using weighted decision-level fusion. The fusion model achieved performance comparable to the speech-only model because emotional information in the TESS dataset is predominantly present in speech.

Overall, this project demonstrates the effectiveness of deep learning approaches for speech emotion recognition and highlights the importance of dataset characteristics in multimodal learning systems. The proposed fusion framework provides a scalable foundation that can be extended to more realistic datasets containing emotionally expressive text, noisy audio conditions, and additional modalities for improved real-world emotion recognition.